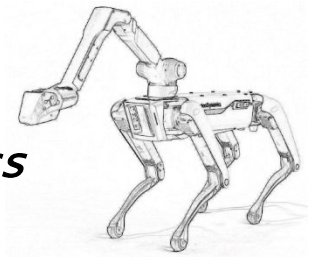


EE 587 Introduction to Robotics

Homework #2

(Given: Nov 05, Due: Nov 22 23:55)



General Info

- The homeworks are for individual study and are for gaining a better understanding of the course topics. Submit a homework that is your own effort.
- In general, use abbreviations for trigonometric functions. i.e., $s\theta = \sin(\theta)$, $c\theta = \cos(\theta)$ etc.
- If you give the coordinates of a vector or a point as an answer, make sure you specify in which frame it is represented. E.g. p^A specified the point resolved (represented) in frame $\{A\}$.
- The programming questions will use Python3 and the “Robotics Toolbox for Python” by P. Corke. This toolbox also uses a sub-toolbox called “Spatial Math Toolbox” that defines the $SO(2)$, $SE(2)$, $SO(3)$ and $SE(3)$ data types with many utility functions. The links to these toolboxes are in oduclass. There are also additional versions of the functions that operate on python arrays.
- In analytic calculations, leave the square roots in analytic form rather than calculating their decimal values.
- You may manually make some drawings if you wish. In this case, please be neat and clear, scan the final drawing and insert it into your homework document.

Questions

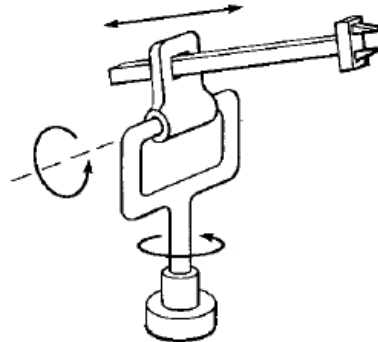
1. Consider the rotation matrix $R \in SO(3)$ given by

$$R = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{\sqrt{2}} & \frac{1}{2} \end{bmatrix}$$

- a. Determine a unit vector w that defines the axis of rotation and the angle θ (in degrees) of rotation.
 - b. Is the pair (w, θ) unique?
 - c. Determine the Euler Parameters $\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4$ of R .
2. In our case, for a rotation matrix R , we had $\text{Det}(R) = +1$. Research and explain what is the meaning of a negative determinant. Cite your sources.
 3. For the 3-dimensional rotation about the vector $[2, 3, 4]$ by 0.5 radians, compute an $SO(3)$ rotation matrix R using
 - a. python the matrix exponential functions `scipy.linalg.expm` and `texp`,
 - b. Rodrigues’ rotation formula (code this yourself), and

- c. the toolbox function `angvec2tr`. Verify that you have the same result in each case.
- d. Compute the equivalent unit quaternion.

4. Consider the following RRP manipulator (from J.J. Craig, “Introduction to Robotics”)



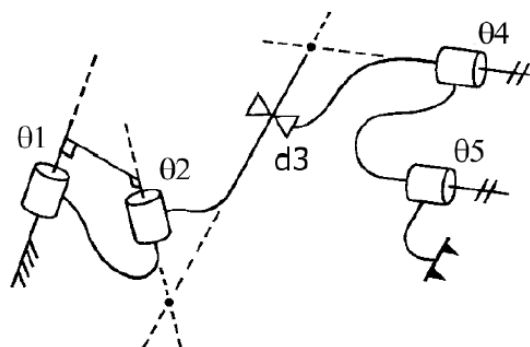
- a. Sketch a manipulator schematic diagram for this manipulator. (**Note:** A “schematic” is composed of only frame origins, frame axes, mutual orthogonal, distances and angles. See lecture slide 20 on fwd. kinematics and think without the actual link bodies shown.)

On your schematic, assign and show link coordinate frames $\{0\}$ to $\{3\}$ using Denavit-Hartenberg convention of the lecture.

- b. Determine the DH parameters and fill in the following table with your values and indicate all nonzero values on your schematic.

Link	a_i	α_i	d_i	θ_i
1				
2				
3				

5. Consider again the 5DOF “RRPRR” manipulator described below. (from J.J. Craig, “Introduction to Robotics”)



- a. Draw a schematic of this manipulator, with the axes of frames $\{0\}$ through $\{5\}$ labeled. Again indicate all non-zero Denavit-Hartenberg parameters and

the joint variables on your figure. Note: Draw your schematic in the position where, as far as possible, the angles θ_i are close to their zero positions.

- b. Determine all DH parameters and fill in the following table.

Link	a_i	α_i	d_i	θ_i
1				
2				
3				
4				
5				

Programming:

6. Consider the Peter Corke's Robotics Toolbox for python again. In your toolbox installation, study and execute the kinematics tutorial given in <https://github.com/petercorke/robotics-toolbox-python/blob/master/notebooks/kinematics.ipynb>
(Note: There are many other Jupyter notebooks in this folder that may be useful later)
7. In this question, you will use what you have learned in to construct a Puma 560 robot simulation where the robot end-effector will trace a 45° tilted "space circle" within the robot reachable workspace. You need to show an animation of the robot executing this motion by solving for the inverse kinematics using the toolbox. Here is a sketch of possible steps:
 - a. Create and show a Puma 560 robot using the toolbox functions,
 - b. Create a set of 50-100 points on a circle on the xy plane using equations of a circle,
 - c. Transform these points using the spatial math toolbox functions to translate and rotate to a suitable location in the Puma 560 robot workspace.
 - d. Show these points in the robot workspace and adjust your homogeneous transformation to adjust the location. You may also need a "scaling" to grow or shrink the circle so as to fit the robot workspace.
 - e. How can you make sure that all points of the circle are within the reachable workspace?
 - f. Solve the inverse kinematics problem for the sequence of circle points to determine a trajectory for all the joints of the robot. The robot end-effector approach vector can be in the plane or orthogonal to the plane of the circle.
 - g. Using python matplotlib and pyplot functions, plot the 6 joint trajectories versus point index.
 - h. Animate the robot so that the motion is shown.
 - i. Capture the animated motion as a video, upload to YouTube as an unlisted video. Share the link in your report.

Good luck.